

# Preliminary Results of Virtual Reality Rehabilitation Program on Executive Functions of Children with Developmental Coordination Disorder

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## Background

The application of Virtual Reality (VR) in the field of rehabilitation has been widely studied, because it has already proven to be an effective intervention for a variety of physical and cognitive conditions. In our previous study, we evaluated feasibility of a VR-rehabilitation program on the Executive Functions (EFs) of children with Specific Learning Disorders. In this pilot study, we present preliminary results regarding the application of a similar methodology for children with Developmental Coordination Disorder, because of key-role of EFs on regulation, monitor and control of motor behaviour towards a goal.

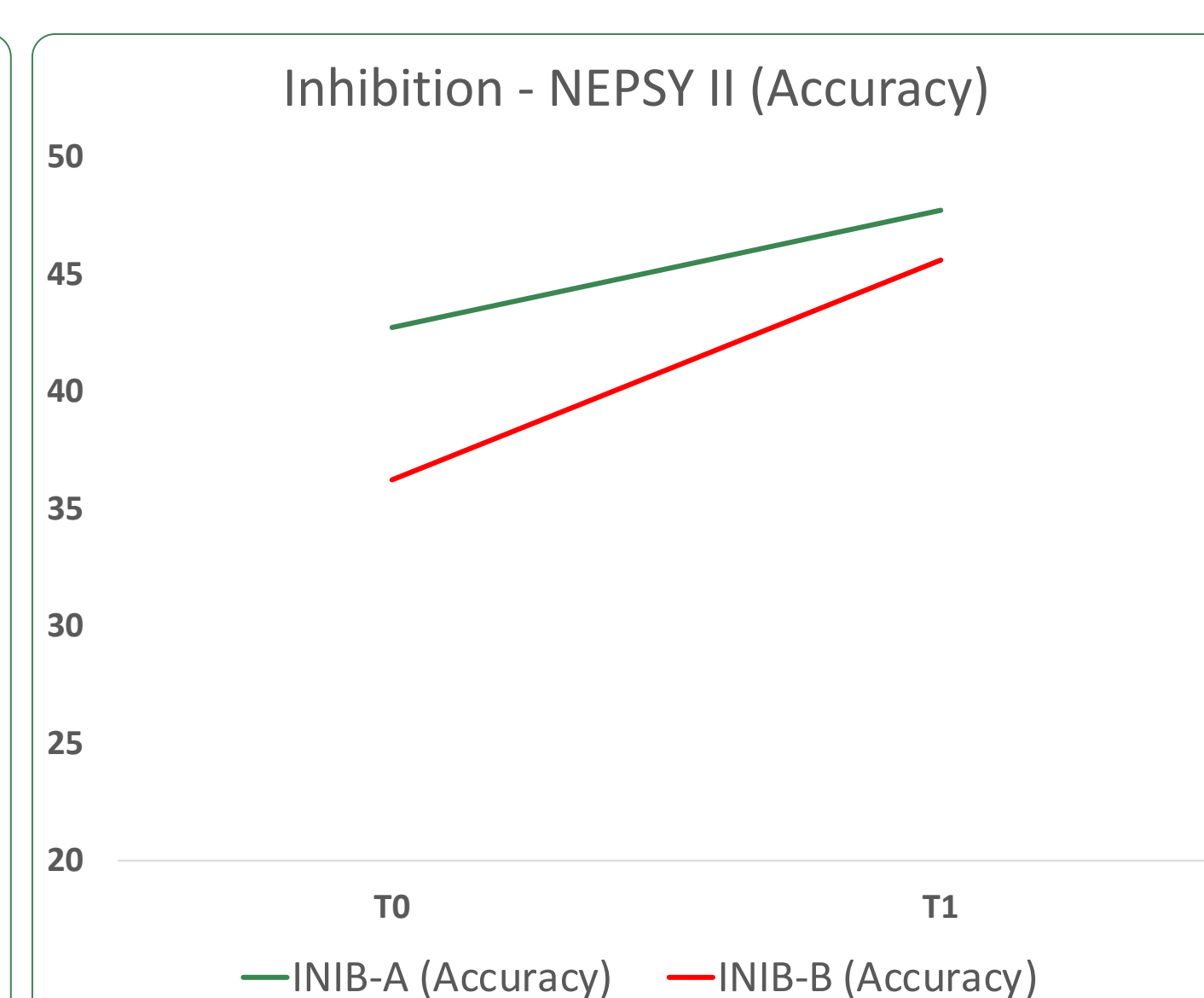
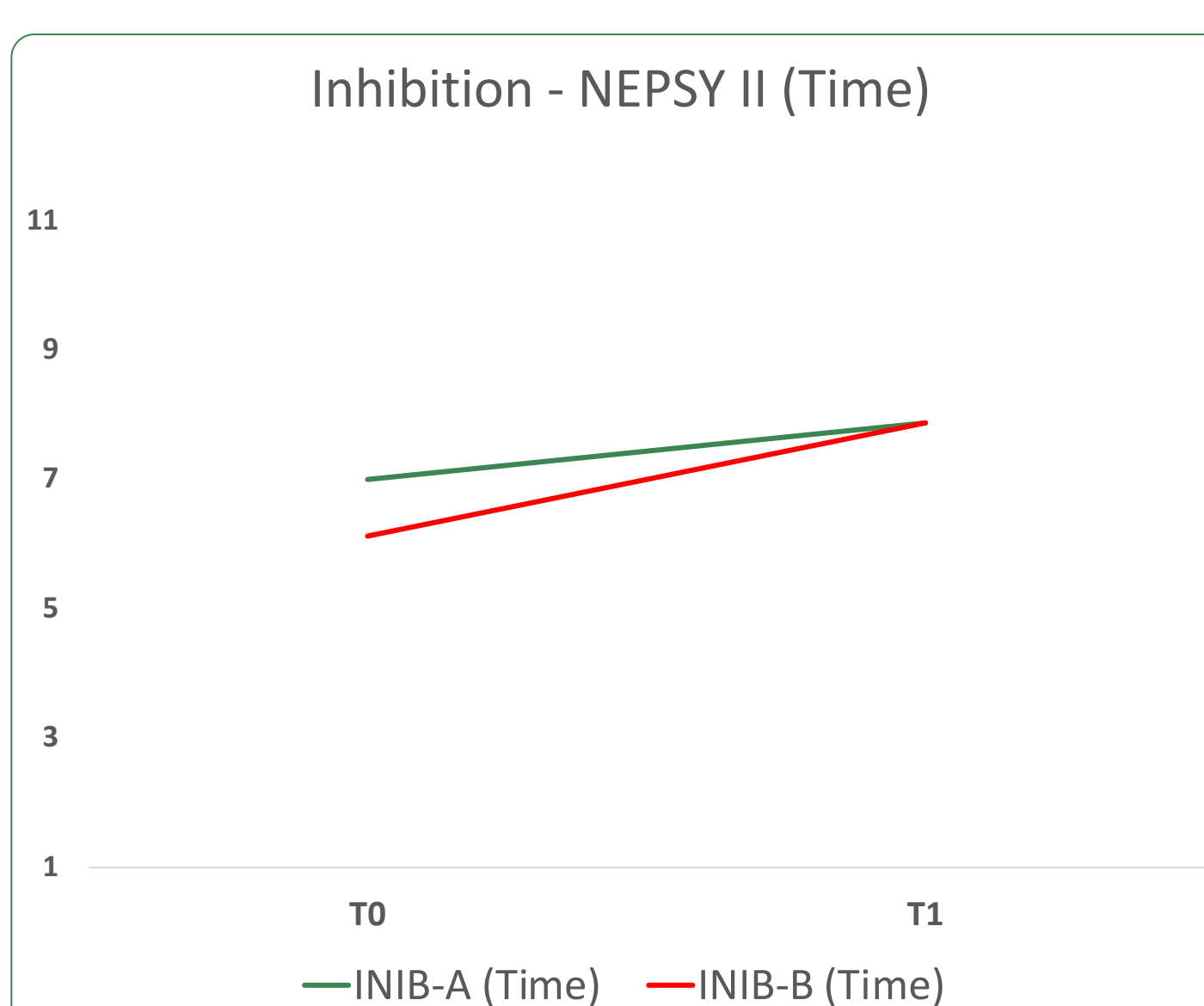


## Methods and participants

Eight school-aged children with diagnosis of DCD participated to the study and performed the VR-training across 16 weeks in the CARE Lab (twice a week), that was designed with appropriate structural measures and ad hoc fittings, to hide the sophisticated technology necessary to allow the child to experience a rehabilitative setting with recreational and semi-immersive features. Children were evaluated before and after the VR-training in visual attention, inhibition, and planning. The rehabilitation programs were customized according to clinical needs and/or single patient's characteristics, proposing different games with variable complexity levels.

## Results

All participants successfully completed the 100% of training sessions. Non-parametric analysis (Mann-Whitney U Test) showed a significant increase of scores in visual attention and inhibition abilities after the VR-based intervention.



|             |             |
|-------------|-------------|
| N° Patients | 8           |
| Sex         | M: 5 / F: 3 |
| Mean IQ     | 93,38       |
| Mean Age    | 8           |

| Evaluation Test                  | Pre   | Post  |
|----------------------------------|-------|-------|
| Visual Attention (NEPSY-II)      | 9,25  | 10,5  |
| Inhibition A Accuracy (NEPSY-II) | 42,75 | 47,75 |
| Inhibition A Time (NEPSY-II)     | 7     | 7,88  |
| Inhibition B Accuracy (NEPSY-II) | 36,25 | 45,63 |
| Inhibition B Time (NEPSY-II)     | 6,13  | 7,88  |
| Maze (WISC-III)                  | 8,5   | 9,75  |

## Conclusion

These findings provide important inputs for the development of new innovative rehabilitation interventions based on Virtual Reality for children with DCD that must be founded in ecological and evidence-based approaches.

## References

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